

RECOMMISSIONING OF THE UNIVERSITY OF HAWAI‘I LINAC AND FREE ELECTRON LASER

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Abstract

The electron beam linear accelerator (linac) at the free-electron-laser (FEL) laboratory of the University of Hawai‘i at Mānoa, originally developed by Prof. John Madey, has undergone recommissioning. The S-band linac delivers 45 MeV electron beams with 170 mA pulse current and 4–8 μ s pulse duration to drive an infrared FEL oscillator. Recent efforts include restoration of the microwave thermionic gun with a new LaB₆ cathode, upgraded vacuum and RF systems, and development of a Python-based beam dynamics model to recover operational magnet settings and optimize beam transport. These upgrades address key technical challenges to restart the accelerator and pave the way for future FEL experiments, including coherent pulse shaping and inverse Compton scattering x-ray generation.

INTRODUCTION

The University of Hawai‘i Free Electron Laser (UH FEL) facility comprises a compact S-band linear accelerator and an infrared FEL system originally developed as the Mark-III FEL [1]. Transferred from Duke University and recommissioned in Hawai‘i under the leadership of Prof. John Madey, the facility entered standby in 2016 following his passing and the pandemic period. Current efforts described in this paper aim to restore beam and FEL operation, including inverse Compton scattering (ICS) experiments [2].

The injector consists of a thermionic LaB₆ cathode in a single-cell 2856 MHz microwave gun operating in the TM₀₁₀ mode. The beam is compressed and energy-filtered by an Enge alpha magnet before injection into a traveling-wave linac, which accelerates the beam to 35–45 MeV. A feed-forward RF stabilization system minimizes phase and amplitude jitter [3].

A diagnostic chicane downstream of the linac provides beam focusing, emittance measurements, and matching into the undulator. It includes five split-bend dipoles, quadrupole lenses, stripline BPMs, and OTR screens.

The undulator is a planar, permanent-magnet device (2.3 cm period) embedded in a laser oscillator cavity. The FEL operates in the mid-IR (2.5–3.5 μ m) and is designed to reach a peak power of 2.6 MW [2]. Previous ICS experiments at UH demonstrated production of ~10 keV x-rays via single-pass interactions of FEL pulses and electron bunches, yielding 3×10^5 photons/s.

This paper presents recommissioning results: vacuum restoration in the gun sector, fabrication of a new LaB₆ cathode, RF amplifier upgrades, and beam dynamics modeling for FEL and ICS operation.

RECOMMISSIONING ACTIVITIES

The main activities leading to the recommissioning of the UH linac and FEL are described here in four subsections.

Vacuum System Restoration

Following several years of inactivity, multiple vacuum sectors had been left vented. Upon restarting the systems, nominal pressures were quickly re-established in the linac and diagnostic chicane. In contrast, both the microwave gun and the spectrometer regions showed signs of leakage. While the spectrometer vacuum was promptly restored after sealing a localized leak, recovering the gun sector required significantly more time and a substantial system upgrade. After removal of the thermionic cathode and reactivation of the pumping system, pressure in the gun initially reached only $4\text{--}5 \times 10^{-7}$ Torr and deteriorated further with time. This sector had relied on two 30 L/s ion pumps, which had become inadequate for maintaining vacuum due to overheating and limited effectiveness in removing nitrogen, a dominant residual gas in case of leaks in the system.

Using a helium leak detector, several leaks were located and sealed. These repairs, combined with the addition of a new 250 L/s turbomolecular pump backed by a dry scroll pump, substantially improved vacuum performance. The turbo pump also enabled in-situ baking of the microwave gun, which further reduced outgassing rates.

As a result of these interventions, the base pressure in the gun sector was restored to $2\text{--}3 \times 10^{-8}$ Torr—comparable to historical operational conditions. This improvement is essential for extending cathode lifetime and achieving stable emission during long macropulses. While the system does not yet reach ultra-high vacuum (UHV), further upgrades are under consideration. These include the integration of Non-Evaporable Getter (NEG) pumps, which are highly complementary to ion pumps and particularly effective for hydrogen and other light gases.

Cathode Fabrication and Installation

The microwave gun utilizes a LaB₆ single-crystal thermionic cathode. Complete cathode assemblies, including the heating system, are fabricated in-house. The heating element consists of a tapered tungsten coil, designed to concentrate electrical resistance in the coil region and efficiently transfer heat via radiation to the LaB₆ button.

Following assembly, each cathode is placed in a vacuum-sealed glass desiccant chamber for outgassing and conditioning. The electrical power is gradually increased in steps of typically 1 W every 6–12 hours, allowing progressive pumping of the outgassing at thermal equilibrium. Most outgassing is completed by 4–6 W of input power. Voltage

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and current are continuously monitored to track resistance evolution. A target power of 16 W, corresponding to approximately 1600 K, is used as the qualification threshold. A consistent rise in resistance indicates a successful conditioning cycle without shorting of the heating elements due to thermal expansion, as shown in Fig. 1.

Using this procedure, a qualified LaB_6 cathode was recently installed in the RF gun and is ready for beam operations. In parallel, heating simulations are underway to model the thermal performance of various coil geometries and eventually improve the efficiency of the system. These simulations will be benchmarked against temperature measurements using a pyrometer.

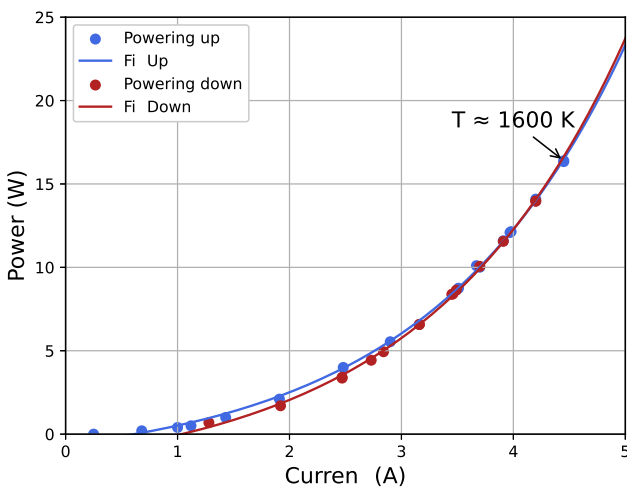


Figure 1: Powering test results of a LaB_6 cathode prototype before installation into the microwave gun.

RF System Upgrade

The S-band RF system for the UH linac was upgraded with the acquisition of a new 1.5 kW solid-state amplifier to replace the original Thomson TH2047 driver klystron. The new amplifier, based on GaN technology, offers high efficiency, fast pulse rise/fall times (< 100 ns), and excellent pulse stability. It will serve as the low-level RF (LLRF) source for the high-power ITT klystron rated at 40 MW peak power that feeds both the linac and the microwave gun.

In preparation for integrating the new amplifier, we performed a comprehensive set of measurements on the RF system. Low-power RF tests using a spectrum analyzer and tracking generator allowed us to map attenuation throughout the system. Losses from the control room to the ITT klystron input were measured at approximately 4.4 dB. The return path from the klystron output to the control room monitoring port showed a total attenuation of 16.4 dB, accounting for fixed attenuators and couplers. These measurements confirm consistent behavior with historical logs and establish a reliable baseline for future power delivery estimates.

High-voltage conditioning of the pulse-forming network (PFN) and the thyatron modulator was successfully completed. Perveance measurements of the ITT klystron indicate

stable operation in the range of 1.8–2.0 μPerv , with no evidence of arcing or vacuum degradation.

The integration of the new 1.5 kW amplifier is ongoing and expected to be the final step before achieving new beams in the linac. Integration of the existing feed-forward stabilization system is also planned to minimize energy spread and arrival-time jitter, critical for stable linac and FEL operation.

Beam Dynamics Simulations

A Python-based model of the UH beamline has been developed to simulate electron transport from the linac, through the diagnostic chicane, and into the FEL undulator. The simulation framework is based on first-order transfer matrix formalism and includes fringe field effects from quadrupoles and dipole edge focusing. The primary objective is to compute beam envelopes, transverse phase space evolution, dispersion, and Twiss parameters at critical locations along the beamline.

The diagnostic chicane features a Chasman-Green Double Bend Achromat (DBA) configuration adapted to a linear geometry [4]. This DBA-inspired transport system is composed of alternating sections with dispersion-generating dipoles and dispersion-free focusing sections. Chromaticity quadrupoles are used to correct the horizontal dispersion.

We have performed several batches of simulations assuming a range of energy chirp at the exit of the linac, as this uncertainty remains until the start-to-end model is completed. A parallel effort to model the injector is underway. An example of the transport from the exit of the linac to the FEL undulator, while assuring ICS operation at the interaction point ($z = 7.11$ m), is displayed in Fig. 2. These simulations assume a normalized emittance of $\varepsilon_n = 8\pi \cdot \text{mm} \cdot \text{mrad}$ and nominal energy of 40 MeV. A summary of the beam parameters derived from the code at key locations of the transport line is displayed in Table 1. At the ICS Interaction Point (ICS IP), the code predicts a focused beam waist with 95% of the beam enclosed within a transverse acceptance of 100 μm -radius, as shown in Fig. 3.

Table 1: Beam Parameters of the UHM Linac

Parameter	Value	Units
Exit of the linac		
Beam Energy	35–45	MeV
Energy Spread	0.5	%
Normalized Emittance	8	$\pi \cdot \text{mm} \cdot \text{mrad}$
Bunch duration	2	ps
ICS interaction point		
Twiss parameter β_x	4.0	mm
Twiss parameter β_y	27	mm
Twiss parameters α_x, α_y	0	
FEL undulator entrance		
Twiss parameter β_x	1.267	m
Twiss parameter β_y	0.242	m
Twiss parameter α_x	0.560	
Twiss parameter α_y	0	

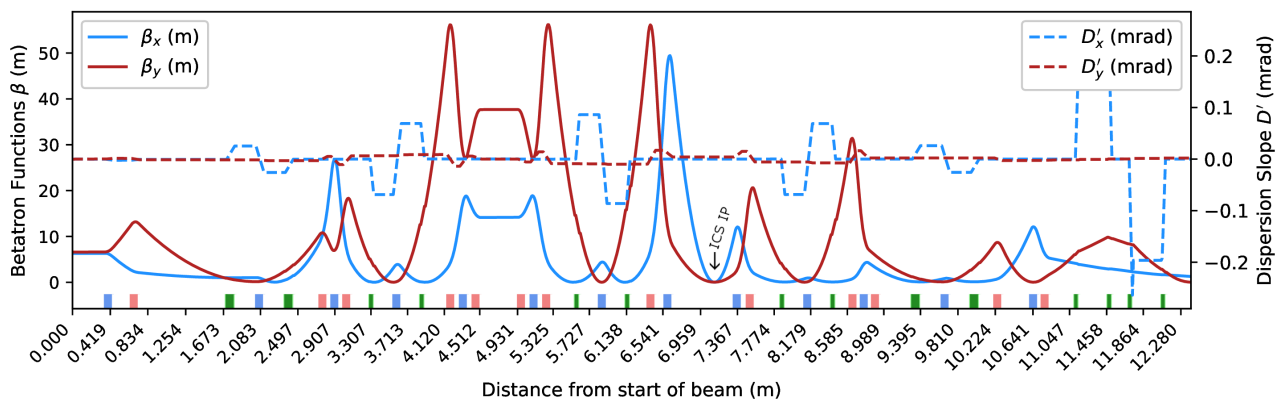


Figure 2: Beam transport simulation from the end of the linac (left, $z = 0$ m) to the start of the FEL undulator (right, $z = 12.38$ m) showing the evolution of the Beta and dispersion prime functions. The patches at the bottom of the figures represent the beamline elements: focusing (blue), defocusing (red) quadrupoles and dipoles (green).

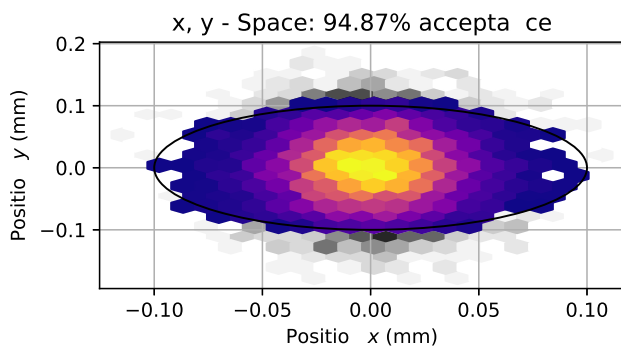


Figure 3: Simulated transverse beam distribution at the ICS interaction point. A transverse acceptance of radius $100 \mu\text{m}$ is drawn to display the beam particles included.

At the FEL undulator entrance, the beam matching in the vertical plane requires a waist and a beta function such that $\beta_{y,m} = \gamma / (K \cdot k_u) = 0.242$ m, where $K = 1.2$ is the undulator parameter, $\lambda_u = 2.3$ cm is the undulator period (with $k_u = 2\pi / \lambda_u$), and γ is the Lorentz factor for a 40 MeV electron beam. In the horizontal plane, the optimal beta function for lasing is $\beta_{x,0} = Z_R \epsilon_r / \epsilon_x = 1.267$ m, with $\epsilon_r = \lambda / (4\pi)$ the FEL radiation emittance at a wavelength $\lambda = 3.23 \mu\text{m}$ and $Z_R = 498.1$ mm is the Rayleigh length, approximately half the undulator length [5]. The beam focusing in the horizontal plane is chosen to reach a waist at the middle of the undulator (Table 1).

This simulation tool was developed in response to the loss of historical magnet settings due to a failure of the historical database system. In addition to finding the operational values of the magnets, it became a flexible and extensible framework that is being expanded to incorporate injector and linac modeling and will support benchmarking against higher-order tracking codes. This code will also facilitate future integration of physics-informed machine learning and optimization techniques for tuning, thus maximizing the throughput of the facility.

DISCUSSION AND OUTLOOK

With the installation of a new LaB_6 thermionic cathode and substantial upgrades to the vacuum and RF systems, the UH FEL facility is nearing readiness for renewed beam operation. The final commissioning of the new solid-state RF pre-amplifier and full RF chain is the last major milestone before restoring the electron beam.

Looking ahead, scientific efforts will focus on leveraging the FEL's mid-infrared output for next-generation light source research. Simulations and design studies are underway to explore optical pulse manipulation inside the FEL oscillator using a Michelson interferometer inserted into the optical cavity. In parallel, the FEL oscillator is being simulated to study the effects of cavity desynchronization. These studies reveal the emergence of post-saturation super-radiance and indicate optimal cavity detuning conditions for maximizing output power and spectral purity [5].

In the near term, facility efforts will focus on beam commissioning, validation of the simulation predictions, and development of precision beam diagnostics. Over the longer term, our goal is to establish a university-scale research center for FEL science and accelerator physics, enabled by improved beam quality, FEL coherence, and optical cavity control. These advances not only extend the legacy of the UH FEL but also open new venues for compact light source research and applications.

ACKNOWLEDGEMENTS

The authors acknowledge Prof. Peter Gorham, Dr. Mike Hadmack and scientific instrument physicist Christian Miki for initiating the recommissioning activities on the linac, as well as undergrad students Daniel Criss and Christian Komo for their work on the cathode fabrication and beam dynamics simulations, respectively. This work is supported by start-up funds provided by the University of Hawai'i at Mānoa to N. Bidault, and by the U.S. Department of Energy, Office of Science, Office of Basic Energy Sciences, under Contract No. DE-SC0025583.

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